# Walter Gnerre

US Citizen | wgner001@fiu.edu | linkedin.com/in/walter-gnerre | Miami, FL | (954) 651-1880

## EDUCATION & HONORS

**Florida International University, Miami, FL** Expected April 2028
Bachelor of Science, Mechanical Engineering | GPA 3.3
**Honors:** OURS Undergraduate Research Scholar (2026–27) | Boeing Honorable Mention, Boeing Week at FIU (2026) | Dean's List, Spring 2026

## SKILLS

**Software:** Python, C++, SolidWorks, Siemens NX, Fusion 360, KiCad, ANSYS, NASA CEA, Revit, AutoCAD, Civil 3D, Arduino IDE, GitHub, MS Office
**Technical:** Spacecraft Systems Engineering, Propulsion Testing, Solar Simulator Design, Thermal-Vacuum Test Planning, Requirements & Budgets, Composite Fabrication, Embedded Systems

## EXPERIENCE

**Undergraduate Research Assistant | FIU College of Engineering** Miami, FL | Jan 2026 – Present
*Solar Thermal Evaporative Propulsion (STEP) — PIs: Dr. Alicia Boymelgreen, Dr. Jarrod Schiffbauer*
- Own design of a vacuum-compatible LED solar simulator targeting 1361 W/m² (AM0) over a 10 × 10 cm plane, for a 1-week test in a NASA Kennedy Space Center dirty vacuum chamber (April 2027)
- Produced the system-level diagram: 2 chamber feedthroughs (power, signal), a conduction-only heat path (MCPCB → aluminum baseplate), and photodiode and thermocouple feedback
- Fabricate carbon-ceramic evaporators by hand and authored the lab's repeatable fabrication procedure
- Ran 15+ evaporation tests from dark to 1-sun, measuring a 23× rise in rate (0.016 to 0.371 μg/s) that supported a ~1 mN thrust estimate at 5-sun

**Undergraduate Intern | Space Dynamics Laboratory** Albuquerque, NM | May 2026 – Jul 2026
*AFRL University Nanosatellite Program — Mission Concept*
- Conceptualized an in-situ method to measure ~1 mN of thrust: offset the thruster from the center of mass so an IMU and reaction wheel capture the induced torque
- Architected the software interface for a 3U CubeSat, mapping data and power connections across 5 subsystems (ADCS, EPS, COMM, TCS, PAY) to the onboard computer, and wrote the command and data handling (CDH) requirements
- Carried the mission concept, objectives, and success criteria through 3 design reviews (MCR, SRR, PDR) before NASA, AFRL, USSF, and industry reviewers

**Payload Lead | FIU SEDS — IREC 2027 / Spaceport America Cup** Miami, FL | Sep 2026 – Present
*SDL Payload Challenge*
- Lead 10 students across 2 subteams (Research; Testing & Integration) and 2 FIU faculty labs building a 3U-form-factor microgravity payload
- Conceptualized a payload studying capillary-driven droplet ejection of passive and self-propelled (Janus particle) colloids with a 9-capillary matrix (3 fluids × 3 replicates)
- Assessed 1–5% $H_2O_2$ against IREC hazard rules, identifying ESRA's non-toxic classification, requesting official confirmation, and drafting a handling, storage, and PPE procedure

**Lead Systems Engineer | NASA L'SPACE Mission Concept Academy** Remote | Jan 2026 – May 2026
- Led an 8-member cross-disciplinary team designing a two-stage lunar rover to prospect for volatiles in the Moon's south-pole permanently shadowed regions
- Absorbed a 23% mass budget cut (150 to 115 kg) with a data-driven trade study across subsystems, guiding the team to a consensus design

## PROJECTS

**Autonomous Robot Vehicle** | *ESP32, C++, Arduino IDE* Summer 2026
- Built a 2-motor DC robot car on an ESP32 and L293D driver; wrote C++ firmware for autonomous 4-direction navigation with no tethered computer

**NAR Level 1 High-Power Rocket** | *Kit build, fin redesign* Fall 2026
- Designed and built a high-power rocket (Fall 2025), modifying fin geometry for improved stability; flew it to earn NAR Level 1 certification (Fall 2026)

**Level 2 Composite Airframe** | *Fiberglass/epoxy, 3D printing* Spring 2026
- Replaced a COTS airframe with a custom fiberglass/epoxy airframe laid up in-house over a 3D-printed support structure to increase rigidity

**Raptor V3 Engine Model** | *SolidWorks* Spring 2026
- Modeled a simplified Raptor V3 (Starship) engine from public data, replicating $CH_4$ inducer shaft rotation

## CERTIFICATIONS

National Association of Rocketry (NAR) Level 1 High-Power Rocket Certification Fall 2026
NASA L'SPACE Mission Concept Academy — Certificate of Completion Spring 2026
*Skill badges: Systems, Requirements, Risk Management, Project Management, Heat Transfer, Siemens NX CAD, Teaming*